

General Knowledge

Art & Culture

STUDY NOTES

Modern Architecture



MODERN ARCHITECTURE



Modern Indian Architecture

Modern Indian architecture was heavily influenced by **European colonists**, who introduced a variety of **architectural styles**. Their arrival coincided with the **decline of the Mughal empire**, sparking a **power struggle** among the **Portuguese, French, Dutch, Danish, and British** colonizers, lasting until **1947**. Each European power left a distinct **architectural imprint** on the Indian landscape.

Portuguese Influence

- The **Portuguese** introduced the **Iberian architectural style**.
- Initially, they built **trading terminals** and **warehouses** along the coast, which later transformed into **fortified cities**.
- They also introduced **patio homes** and the **Baroque style** to **symbolize the Church's strength**.
- **Baroque style** is marked by **complex, dramatic, and theatrical designs**, often using **clashing colours** for visual impact.

Portuguese Structures in India

Name of Structure	Location	Year Built	Architectural Style	Key Features
Sé Cathedral	Goa	1619 AD	Late-Gothic Portuguese Style	Houses the Golden Bell , the largest bell in Goa.

Basilica of Bom Jesus	Goa	1604 AD	Baroque Style	UNESCO World Heritage Site; contains relics of St. Francis Xavier .
Castella de Aguada	Mumbai	-	Portuguese Fortification	Controlled access to Mahim Bay .
St. Paul's Church	Diu	1610 AD	Baroque Style	Known for its decorative interiors .
Diu Fort	Diu Island	1535 AD	Portuguese Fortification	Includes a lighthouse, cannons , and three churches.
St. Anne's Church	Talaulim, Goa	1695 AD	Baroque Style	Elaborate design with dramatic architectural elements .

French Influence

The **French influence** in Indian architecture introduced new concepts of **urban planning** and **scientific architectural designs**. Their architectural legacy is evident in the **French colonies of Puducherry, Chandernagore (Chandannagar, West Bengal)**, and several other coastal cities across India.

Urban Planning and Cartesian Grid Layouts

- The **French brought the concept of urban city planning**, focusing on **organized city layouts**.
- They implemented **Cartesian grid systems**, a geometric layout based on **straight streets** intersecting at right angles.
- These principles of city planning were evident in the development of **French colonies** such as **Puducherry** and **Chandernagore**.
- Their structured design not only made the cities **more organized** but also served as a **display of authority and control**.

Massive and Minimalist Structures

- The **French colonizers** constructed **massive buildings** to **assert their power and influence**.
- They also introduced **faceless architecture**, characterized by **simple facades** without decorative elements, focusing more on **functionality** rather than embellishment.
- This architectural style can be seen as an early example of **modern minimalist architecture**, where **design simplicity** and **plain surfaces** were prioritized over ornamentation.

Key French Coastal Cities in India

- The French established several **coastal cities** in India, including:
 - **Mahe (Kerala)**
 - **Karaikal (Tamil Nadu)**
 - **Yanam (Andhra Pradesh)**

Prominent French Colonial Structures

- **Sacred Heart Church (Puducherry)**
 - A prominent example of **French Gothic architecture** with tall spires and stained-glass windows.
- **Sacred Heart Church (Chandannagar, West Bengal)**
 - Another example of **French colonial architecture**, emphasizing **simplicity and grandeur**.

French Colonial Cities and Structures in India

City/Structure	State/Location	Key Features
Puducherry	Tamil Nadu	Built with Cartesian grid layout and massive structures.
Chandernagore	West Bengal	Scientific city planning with a focus on faceless facades .

Mahe	Kerala	A coastal town with French colonial architecture .
Karaikal	Tamil Nadu	Exhibits minimalist French architecture .
Yanam	Andhra Pradesh	Reflects French influence in city planning and design.
Sacred Heart Church	Puducherry	French Gothic design with spires and stained glass .
Sacred Heart Church	Chandernagore (Chandannagar)	Example of French colonial architecture with simple facades.

British Influence on Indian Architecture

The **British colonial period** in India introduced significant **architectural innovations**, especially the **Gothic style** of construction. Over time, the **fusion of Gothic architecture with Indian elements** gave rise to the **Indo-Gothic style**.

Post-**1911**, the British began incorporating **Neo-Roman or Neo-Classical styles**, leaving a lasting impact on the architectural landscape of India.

1. Indo-Gothic Style (Victorian Style)

- The **Indo-Gothic style**, also known as the **Victorian style**, represented a unique **combination of Indian, Persian, and Gothic elements**.
- This style became widely popular due to its **large, intricately designed structures** and use of **pointed arches** (in contrast to the curved arches of Indo-Islamic architecture).

Characteristics of Indo-Gothic Style

- **Extraordinarily large and intricate structures** were built, showcasing grandeur.
- **Pointed arches** were used, distinguishing it from earlier architectural styles like Indo-Islamic.
- **Wide windows** became a common feature, improving ventilation and natural light.
- Followed **advanced structural engineering practices** prevalent in Britain.

- **Steel, iron, and poured concrete** were widely used for construction.
- Structures often followed a **cruciform ground plan**.
- Examples include the **Victoria Memorial in Kolkata** and the **Gateway of India in Mumbai**.

Key Examples of Indo-Gothic Architecture

- **Victoria Memorial (Kolkata)**
 - A large marble monument combining Indian and Gothic design elements.
- **Gateway of India (Mumbai)**
 - Built to commemorate King George V's visit, reflecting Indo-Gothic aesthetics.

Comparison: Iberian vs. Gothic Architecture

Basis	Iberian Architecture	Gothic Architecture (British Influence)
Material Used	Used brick as the primary building material, with roofs and stairwells made of wood .	Used red sandstone and coarse limestone as primary materials.
Structural Variations	Retained Western customs without significant structural changes.	Developed Indo-Gothic style by incorporating Indian elements.

2. Neo-Roman or Neo-Classical Design (Post-1911)

- After **World War I**, the British shifted towards **Neo-Roman (Neo-Classical) architecture**, marking a new architectural phase in India.
- The **buildings of New Delhi**, designed by **Edwin Lutyens** and **Herbert Baker**, are notable examples of this style.
- The city came to be known as the "**Rome of Hindustan**" for its architectural grandeur.

Characteristics of Neo-Roman or Neo-Classical Style

- **Unremarkable structures** with no distinguishing features became common.

- This style was a **mash-up of multiple architectural styles**, creating **crowded designs** that limited artistic expression.
- **Simplicity, modernism, and usability** were compromised due to the **composite nature** of the buildings.
- Special emphasis was placed on **circular structures**.
- **Eastern motifs** were excessively used to implement **Western architectural principles**.

Key Examples of Neo-Roman Architecture

- **Supreme Court of India**
 - Features an **upturned dome**, symbolizing the fusion of Indian and Western elements.
- **Rashtrapati Bhavan (New Delhi)**
 - A grand structure with an **upturned dome**, showcasing British architectural influence.

Architecture in the Post-Independence Era

Following **India's independence in 1947**, two distinct architectural movements emerged: **Revivalist and Modernist schools**. Both sought to reshape Indian architecture but struggled to shed the **influence of colonial styles**. As a result, the true essence of **India's traditional architecture** declined, leaving an imprint of Western principles on the nation's construction landscape. Several architects and projects during this period attempted to **balance modern architectural needs with cultural relevance**.

Revivalist and Modernist Schools of Architecture

- **Revivalist School**: Aimed to **revive traditional Indian architecture** by incorporating **historic design elements** into modern structures.
- **Modernist School**: Focused on **contemporary architectural principles**, emphasizing **functionality, scientific engineering, and modern materials**.
- Both movements, however, struggled with a **colonial hangover**, resulting in designs that lacked originality and continuity with **India's rich architectural heritage**.

Le Corbusier and the Creation of Chandigarh

- The **Punjab government** commissioned **Le Corbusier**, a renowned **French architect**, to design **Chandigarh**, India's first **planned city**.
- **Le Corbusier's design** reflected **modernist ideals** with a focus on **urban functionality** and **scientific planning**.
 - The city layout included **sector-based divisions**, each with defined spaces for residential, commercial, and recreational activities.
 - **Modern materials** like **concrete and steel** were used extensively to align with the functionalist approach of Modernism.
- While Chandigarh is celebrated for its **modernist planning**, it was criticized for lacking a **connection to Indian architectural traditions**.
- Today, Chandigarh stands as an example of **urban modernism**, but its heavy reliance on **Western design philosophies** highlights the struggle to reconcile modernity with local culture.

Laurie Baker – Architect of the Poor

- **Laurie Baker** revolutionized housing design in **Kerala**, becoming known as the **"Architect of the Poor."**
- His architectural approach focused on **affordable, sustainable, and eco-friendly construction**, using **locally available materials** to reduce costs.
- Baker was **nominated for the Pritzker Prize** (known as the **Nobel Prize for Architecture**) in 2006.
- **Key Features of Laurie Baker's Architectural Style:**
 1. **Environmentally friendly buildings** that minimise ecological impact.
 2. **Filler slab construction** to reduce the usage of **cement and steel**.
 3. Emphasis on **ventilation and thermal comfort**, ensuring energy-efficient homes.
 4. Known as the **"Conscience Keeper of Indian Architecture"** for his commitment to socially responsible design.
 5. Referred to as the **"Gandhi of Indian Architecture"** for his focus on simplicity and social welfare.

Charles Correa – Urban Planner and Architect

- **Charles Correa** was a pioneer in **urban design and contemporary architecture**, focusing on the **local needs and sensitivities** of Indian society.
- His work bridged **modern architectural concepts** with **cultural relevance**, ensuring that each project addressed **practical as well as social needs**.
- **Notable Projects by Charles Correa:**
 1. **Madhya Pradesh Assembly Building** – A symbol of modern architecture integrated with Indian identity.
 2. **Mahatma Gandhi Memorial Museum (Ahmedabad)** – Designed to reflect the simplicity and ideals of Gandhi.
 3. **LIC Building (Connaught Place, New Delhi)** – A modern office building located in a historic urban hub.
- In **2006**, Charles Correa received the **Padma Vibhushan Award** for his **contributions to architecture and urban planning**.

C.H. Lutyens

- **Sir Edwin Landseer Lutyens** was a **British architect**, often referred to as **C.H. Lutyens**, who played a significant role in shaping the **architectural landscape of New Delhi**.
- His work in **India's capital city** left an enduring mark, blending **Classical European styles with Mughal elements** to create an iconic architectural identity.

Key Contributions

- **Design of New Delhi:** Lutyens was **commissioned to design the layout of New Delhi** during British colonial rule, working alongside **Herbert Baker**.
- **Rashtrapati Bhavan (Viceroy's House):** One of Lutyens' most iconic works, originally built as the **Viceroy's residence**, now serves as **India's Presidential Palace**.
 - The building blends **Western Classical architecture** with **Indian elements**, such as the **Chhatra (dome-shaped pavilions)**.
 - Features the **upturned dome**, a symbolic fusion of **Eastern and Western architectural styles**.
- **India Gate (All India War Memorial)**

- A monumental structure designed by Lutyens, resembling a **triumphal arch**, built to commemorate the **soldiers of the British Indian Army** who lost their lives in World War I.
- India Gate stands on the **Rajpath**, New Delhi's ceremonial boulevard.
- **Lutyens' Delhi:** The area in New Delhi that houses **government buildings, embassies, and bungalows** is known as **Lutyens' Zone**.
 - His urban planning emphasized **symmetry, large green spaces**, and **well-organized public buildings**.

Post-Independence Architects and Their Contributions

Architect	Key Contributions	Notable Projects	Recognition
Le Corbusier	Designed Chandigarh , India's first planned city. - Introduced sector-based planning with functional zoning. - Focused on concrete structures and modern urbanism.	Chandigarh – A symbol of Modernist architecture in India.	Known for redefining urban planning in India .
Laurie Baker	Introduced eco-friendly construction techniques . - Developed filler slab construction to reduce costs. - Focused on ventilation and thermal comfort. - Known as "Architect of the Poor" and "Gandhi of Indian Architecture."	Revolutionized mass housing in Kerala .	Nominated for the Pritzker Prize (2006) .

Charles Correa	Integrated modern architecture with local needs. - Pioneered urban design with sensitivity to Indian contexts . - Designed important civic and cultural buildings.	- Madhya Pradesh Assembly Building - Mahatma Gandhi Memorial Museum - LIC Building, New Delhi	Awarded Padma Vibhushan (2006) .
-----------------------	--	--	---

Modern Indian Architecture in the Post-Independence Era

Architecture	State/Location	Architect	Year of Completion	Key Features
Chandigarh City	Punjab	Le Corbusier	1953	Sector-based planning, modernist ideals, concrete and steel structures.
Supreme Court of India	New Delhi	Herbert Baker	1958	Neo-Classical style with an upturned dome, reflecting British influence.
Rashtrapati Bhavan	New Delhi	Edwin Lutyens	1931	Blend of classical and Mughal elements, upturned dome, symbolic architecture.
Madhya Pradesh Assembly Building	Bhopal, Madhya Pradesh	Charles Correa	1996	Integrated local elements with modernist architecture.

Mahatma Gandhi Memorial Museum	Ahmedabad, Gujarat	Charles Correa	1963	Designed to reflect the simplicity and ideals of Gandhi.
LIC Building	Connaught Place, Delhi	Charles Correa	1986	Modern office building with a functional and urban focus.
Victoria Memorial	Kolkata, West Bengal	William Emerson	1921	Indo-Gothic design, large marble structure, public museum.
Gateway of India	Mumbai, Maharashtra	George Wittet	1924	Indo-Gothic style, large arch facing the Arabian Sea.
Sabarmati Ashram	Ahmedabad, Gujarat	Charles Correa	1963	Simple design reflecting Gandhi's ideals and lifestyle.
Chhatrapati Shivaji Maharaj Terminus	Mumbai, Maharashtra	Frederick Stevens	1887	Victorian Gothic Revival style, UNESCO World Heritage Site.
IIM Ahmedabad Campus	Ahmedabad, Gujarat	Louis Kahn	1974	Red brick design with bold geometric shapes.
Indian Institute of Science	Bengaluru, Karnataka	C. H. Lutyens	1911	Gothic style with large open spaces and modern research facilities.

Jawahar Kala Kendra	Kerala	Jaipur, Rajasthan	Charles Correa	1992	Inspired by ancient Vastu principles, cultural center with a modern twist.
Baker Houses	Model	Kerala	Laurie Baker	1970s	Affordable eco-friendly housing using local materials and filler slab technique.
Kanchanjunga Apartments		Mumbai, Maharashtra	Charles Correa	1983	High-rise luxury apartments integrating natural ventilation and light.
Lotus Temple		Delhi	Fariborz Sahba	1986	Bahá'í House of Worship with a lotus-inspired design.
ISRO Headquarters		Bengaluru, Karnataka	Charles Correa	1980s	Reflects modern technology and functional design.
Vidhana Soudha		Bengaluru, Karnataka	Kengal Hanumanthaiah	1956	Neo-Dravidian style with elements of Mysore and Indo-Saracenic architecture.
India Habitat Centre		Delhi	Joseph Allen Stein	1993	Modern mixed-use complex with open courtyards and green spaces.

Nehru Centre	Mumbai, Maharashtra	Charles Correa	1980s	Cultural and research centre with a focus on modern architecture.
State Bank of India Building	Mumbai, Maharashtra	Charles Correa	1975	Functionalist design with large glass windows and steel frames.
Tata Theatre	Mumbai, Maharashtra	Philip Johnson	1969	Combines modern design with excellent acoustics.
National Institute of Design (NID)	Ahmedabad, Gujarat	Gautam Sarabhai	1961	Modern design with focus on open spaces and functional areas.
Supreme Court Library	Delhi	Laurie Baker	1980	Eco-friendly structure with natural ventilation and minimal ornamentation.
British Council Building	Delhi	Charles Correa	1992	Inspired by Mughal architecture, with jaali designs and geometric patterns.
Shri Ram Centre for Performing Arts	Delhi	Shiv Nath Prasad	1976	Functional modernist design for performance arts.

Indira Gandhi National Centre for Arts	Delhi	Ranjit Sabikhi	1985	A blend of traditional and modern architectural elements.
Indian Institute of Technology (IIT)	Kanpur, Uttar Pradesh	Achyut Kanvinde	1960	Modernist architecture with open spaces and research facilities.
AIIMS (Main Campus)	Delhi	Mahendra Raj	1956	Modern healthcare facility with open courtyards and natural light.
Tata Institute of Social Sciences	Mumbai, Maharashtra	Achyut Kanvinde	1944	Functional design focusing on academic and social spaces.
NCPA (National Centre for Performing Arts)	Mumbai, Maharashtra	Philip Johnson	1985	Performance venue with a focus on acoustics and modern design.
Rajiv Gandhi International Airport	Hyderabad, Telangana	HOK International	2008	Modern airport with sustainable design features.
Infosys Campus	Bengaluru, Karnataka	Hafeez Contractor	2000	Futuristic design with energy-efficient infrastructure.
DLF Cyber City	Gurugram, Haryana	Hafeez Contractor	2005	Modern business hub with glass

				facades and smart facilities.
Tidel Park	Chennai, Tamil Nadu	C.R. Narayan Rao	2000	One of the largest IT parks in India, with a focus on green building practices.
Amity University Campus	Noida, Uttar Pradesh	Hafeez Contractor	2005	Modern educational complex with smart facilities.
Rashtrapati Niwas	Shimla, Himachal Pradesh	Henry Irwin	1888	Victorian-style summer residence of the President of India.
Indian Parliament House	New Delhi	Herbert Baker	1927	Circular design with large open spaces, inspired by British colonial architecture.
Pragati Maidan	Delhi	Raj Rewal	1972	Exhibition complex with modernist pavilions.
Indira Gandhi Indoor Stadium	Delhi	Sharat Das	1982	Large sports facility with a dome-shaped roof.
ITC Grand Chola Hotel	Chennai, Tamil Nadu	Rajiv Kathpalia	2012	Combines Dravidian architecture with modern amenities.
The Oberoi Udaivilas	Udaipur, Rajasthan	Nimish Patel	2002	Inspired by traditional Rajasthani

				architecture, with modern luxury.
Hyderabad Metro Rail	Hyderabad, Telangana	Hafeez Contractor	2017	Modern public transport system with sustainable infrastructure.
Mumbai International Airport (T2)	Mumbai, Maharashtra	Skidmore, Owings & Merrill	2014	State-of-the-art design with traditional Indian motifs.